Topology-Aware Graph Transformers for detecting voltage loss after distribution network failures

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***Abstract*—This letter proposes a novel Graph-transformer-based method to enforce the physical constraint of network topology in power networks, which requires that buses can only be energised if there exists a path of energised buses and branches to a voltage source. Grid failures and voltage dips are a subject of power system planning, particularly in distribution networks, where planned developments and interconnections are simulated to assess the impact on customer interruptions in reliability analyses. This letter proposes a learning framework that uses bus propagation to predict voltage loss when a network failure interrupts the path to a voltage source. The results show that the model can replace power system software used in simulations with xx% accuracy and yy% less time.**

***Index Terms*—Grid foundation models, Graph transformers, Reliability, Voltage loss detection, Network topology**

I. Introduction

Bus voltages in power networks are topographically constrained to a physical connection of that bus to a voltage source. The occurrence of failures in distribution networks creates voltage islands with no physical path to the grid supply point. Customers interrupted in these islands are a key concern for distribution network operators (DNOs), who design and operate networks to minimise the number and frequency of customer interruptions. Reliability assessments are conducted by DNOs in system planning to identify customers who are interrupted after network failures are simulated stochastically in the network, particularly for reliability improvement investments and the connection of distributed generators and loads [].

Grid foundation models are large artificial intelligence models trained on broad data, using self-supervision at scale, based on transformer architectures that can be fine-tuned to specific power system applications with minimal labelled data [1]. During training, foundation models can predict masked features, and application-specific loss functions penalise these predictions across a series of state-reconstruction tasks, enabling the model to learn the physical characteristics of the grid. The graph transformer architecture employed in [2] predicted bus voltages and currents using a loss function that penalises voltage and current residuals computed from the power balance equation.

This letter proposes a novel propagation-based loss function to enhance a graph transformer's ability to learn the topology of a power network by classifying collapsed bus voltages of nodes that are not physically connected to a voltage source. The loss function is implemented as a reconstruction task that penalises the Graph Transformer's weights across the model's layers during training. The model learns to predict bus availability using a bus-status logit based on the network topology after a contingency is passed, while enforcing the network's physical topology on predicted voltages and other power-system quantities from other tasks. The Graph Transformer used in this study has 4 attention heads that predict the bus voltage magnitude (V_m), bus voltage angle (V_a), bus power demand (P_d), reactive power demand (Q_d), and the bus status logit (z).

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II. Input and Output Data

During the forward pass at each layer of the model, the loss function requires the node and edge features specified in Table I. The values are loaded as tensor columns with power quantities normalised to p.u. on the system's MVA base. The outputs of the loss function are the parameters predicted by the transformer's prediction heads $[V_m, V_a, P_g, Q_g, z]$.

TABLE I
Data input and output tensors

Input/Output	Parameter	Notation
Bus inputs	PV-bus indicator	p_i
	reference / slack bus indicator	r_i
	bus ON/OFF status pre-contingency	s_i
	bus contingency to be applied	c_i
Bus target	voltage magnitude (p.u.)	V_i
	post-contingency energised flag	y_i^{target}
Bus output	predicted voltage magnitude (p.u.)	$\hat{V}_i$
	bus status logit	z_i
Edge inputs	branch ON/OFF status pre-contingency	s_e
	branch contingency to be applied	c_e

III. Methodology

A. Derivation of Node reachability

Buses are classified as energised using a reachability label computed via a breadth-first search along the available buses from the available voltage sources. With a set input threshold ($0 \leq \tau \leq 1$), a bus or branch is labelled available (α_i, β_i) if its status is above the threshold and is not directly hit by a contingency. Voltage source buses are also labelled as available (σ_i) if their status exceeds the threshold, as shown in (2). The physical availability of buses and voltage sources can be

probabilistically modelled for the task with the threshold set accordingly.

$$\alpha_i = \mathbf{1}[s_i > \tau] \wedge \neg\mathbf{1}[c_i > \tau] \quad \text{(bus } i \text{ available)}, \tag{1}$$
$$\beta_e = \mathbf{1}[s_e > \tau] \wedge \neg\mathbf{1}[c_e > \tau] \quad \text{(branch } e \text{ available)}. \tag{2}$$
$$\sigma_i = \mathbf{1}[r_i > \tau] \vee \mathbf{1}[p_i > \tau] \quad \text{(Source at bus } i \text{ available)} \tag{3}$$

A seed set of available voltage sources is computed as $S = \{\, i : \alpha_i \wedge \sigma_i \,\}$. If $S = \emptyset$, the function returns $y^{\text{reach}} \equiv 0$, implying that the whole network is in a blackout. β_{ij} denotes the symmetric branch-availability gate between buses i and j.

Starting at $R^{(0)} = S$, a breadth-first search step adds every bus j that is available (α_i) and is connected to at least one available branch (β_{ij}) that is already reachable as in (4) using the Jacobi update method [3] that simultaneously updates all the nodes of the BFS in one sweep. The propagation is restricted to only available buses, $R^\star \subseteq \{i : \alpha_i = 1\}$.

$$R_{k+1} = R_k \cup \Big\{ j \;:\; \alpha_j \;\wedge\; \exists i \in R_k \text{ with } \beta_{ij} \Big\}. \tag{4}$$

The iteration is terminated at the first k for which $R^{(k+1)} = R^{(k)}$, or after N iterations, the number of buses in the network. Reachable buses are labelled with y_i^{reach} as in (5) as the BFS traverses the network through vectorised frontier expansion in Figure 1.

$$y_i^{\text{reach}} = \mathbf{1}[i \in R^\star] \in \{0, 1\} \text{ where } R^\star = \lim_k R^{(k)}. \tag{5}$$

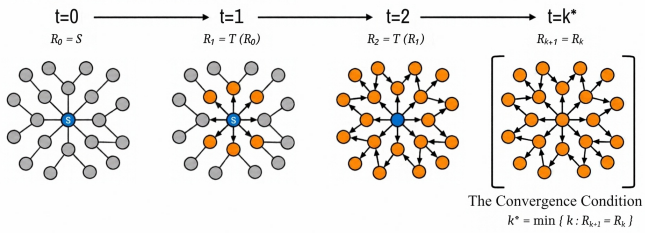


Fig. 1. BFS using the Jacobi update BFS

The buses can then be split into three distinct classes:

1) Unavailable: $\alpha_i = 0$ and $y_i^{\text{reach}} = 0$;
2) Available but physically islanded: $\alpha_i = 1$ but $y_i^{\text{reach}} = 0$
3) Available and energised: $\alpha_i = 1$ and $y_i^{\text{reach}} = 1$

The reachability label y_i^{reach} is used to classify the energized (n_{on}) and de-energised buses (n_{off}) according to (2).

1) Energised buses: $n_{\text{on}} = \sum_i \mathbf{1}\big[y_i^{\text{reach}} \geq 0.5\big]$
2) De-energised buses: $n_{\text{off}} = \sum_i \mathbf{1}\big[y_i^{\text{reach}} < 0.5\big]$

A weight is derived for de-energised buses according to (6) and used as a label for the calculated losses.

$$w_{\text{off}} = \begin{cases} n_{\text{on}}/n_{\text{off}} & n_{\text{off}} > 0, \\ 1 & n_{\text{off}} = 0 \end{cases} \tag{6}$$

The bus weight for each of the bus classes is calculated as the product of the de-energised bus weight and the reachability label for the respective class, as in (7).

$$\omega_i^{\text{cls}} = \begin{cases} w_{\text{off}} & y_i^{\text{reach}} < 0.5 \\ 1 & \text{otherwise} \end{cases}, \tag{7}$$

The graph data is expected to have a majority of energised buses, and the learning objective is towards learning to predict de-energised buses so a pre-set topology confidence factor γ can be applied as in (8) so that $\kappa_i = e^{-\gamma}$ on islanded buses and 1 on energised buses. When γ is set to 0, the weight is clamped to 1 for islanded buses, while a positive γ reduces this weight below 1, so physically islanded buses receive a gradient $e^{-\gamma}$ times smaller than that of de-energised buses of the same class, thereby reducing the emphasis on learning the topology of islanded buses.

$$\kappa_i = \exp\big(-\gamma\,|\alpha_i - y_i^{\text{reach}}|\big) \text{ and } \omega_i = \kappa_i\,\omega_i^{\text{cls}} \tag{8}$$

Another weight for the labelled bus status ν_i is computed with no confidence factor as in (9).

$$\nu_i = \begin{cases} w_{\text{off}} & y_i^{\text{target}} < 0.5 \\ 1 & \text{otherwise} \end{cases}, \tag{9}$$

B. Loss formulation

A binary cross-entropy (BCE) loss is calculated over the bus status logit (z_i) with predicted probability $\hat{p}_i$ and activation function $\sigma(z)$ to classify the masked features. This binary classifier measures the divergence between target labels for the different bus classes and the predicted logit for the bus status. Considering the activation function in (10), the BCE loss is calculated according to (11), which can further be reduced to (12).

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{10}$$
$$\ell(z, y) = -\big[y \log \sigma(z) + (1 - y) \log(1 - \sigma(z))\big] \tag{11}$$
$$= \max(z, 0) - zy + \log\big(1 + e^{-|z|}\big) \tag{12}$$

The primary loss is calculated as the weighted mean of the BCE over the predicted bus-status logit (z) across all buses in the graph, as given by (13).

$$\mathcal{L}_{\text{reach}} = \frac{\sum_{i=1}^{N} \omega_i\, \ell\big(z_i, y_i^{\text{reach}}\big)}{\sum_{i=1}^{N} \omega_i} \tag{13}$$

A secondary anchor loss is calculated for the labelled bus status as a weighted mean over all the buses according to (14).

$$\mathcal{L}_{\text{anc}} = \frac{\sum_{i=1}^{N} \nu_i\, \ell\big(z_i, y_i^{\text{target}}\big)}{\sum_{i=1}^{N} \nu_i} \tag{14}$$

Consider n de-energised buses in n_{off}, a third loss is calculated for predictions of de-energised buses as in (15) to drive the predicted voltages towards 0. A weight λ_l can be set to further drive the model to correctly label de-energised buses, as they are the minority in the dataset.

$$\mathcal{L}_{\text{off}} = \frac{1}{n} \sum_{i \in n_{\text{off}}} \hat{V}_i{}^2 \; + \; \lambda_1 \frac{1}{n} \sum_{i \in n_{\text{off}}} |\hat{V}_i| \tag{15}$$

Finally, each energised bus is weighted by the model's current stop gradient with respect to the probability of predicting the status logit in the primary loss $\text{sg}(\hat{p}_i)$, so that buses that the model is confident are energised contribute more to the voltage regression. The soft confidence weight η_i is calculated according to (16) so that the voltage prediction head fits buses that the bus status logit head is already confident in.

$$\eta_i = \mathbf{1}[i \in n_{\text{off}}] \cdot \text{sg}(\hat{p}_i) \tag{16}$$

A weighted Mean Squared Error (MSE) loss in $\hat{V}$ for the energised buses is then calculated, weighted by the model's current stop gradient (η_i), as in (17).

$$\mathcal{L}_{\text{on}} = \frac{\sum_{i=1}^{N} \eta_i \left(\hat{V}_i - V_i\right)^2}{\sum_{i=1}^{N} \eta_i} \tag{17}$$

$\mathcal{L}_{\text{on}}$ is scale-invariant because it is normalised across all buses and doesn't influence the bus status logit head z_i. This loss guides the model towards minimising the MSE in voltage predictions $\hat{V}_i$, it believes are already energised. This approach accelerates the model's training of physical quantities like voltage once it gains a high confidence in the network topology from pre-training on the primary BCE loss.

The total loss combines all loss terms weighted according to (18). The weights ($0 \leq \lambda \leq 1$) are loss-function parameters set according to the training objective.

$$\mathcal{L}_{\text{VLD}} = \lambda_{\text{reach}} \mathcal{L}_{\text{reach}} + \lambda_{\text{anc}} \mathcal{L}_{\text{anc}} + \lambda_{\text{off}} \mathcal{L}_{\text{off}} + \lambda_{\text{on}} \mathcal{L}_{\text{on}} \tag{18}$$

IV. How the model learns the network's physical topology

The gradients of the loss function $\mathcal{L}_{\text{VLD}}$ are passed to the model through the bus status logit z_i and the voltage magnitude $\hat{V}_i$ as detailed below.

A. Learning the bus status logit

The bus status z_i head is trained purely on the primary and secondary BCE terms $\mathcal{L}_{\text{reach}}$ and $\mathcal{L}_{\text{anc}}$ with no contribution from the voltage terms $\mathcal{L}_{\text{on}}$ and $\mathcal{L}_{\text{off}}$ that influence the predicted voltage $\hat{V}_i$ only. The gradient of z_i can be calculated according to (19).

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{VLD}}}{\partial z_i} &= a_i(\hat{p}_i - y_i^{\text{reach}}) \; + \; b_i(\hat{p}_i - y_i^{\text{target}}) \\ &\text{with } a_i = \lambda_{\text{reach}} \frac{\omega_i}{\sum_i \omega_i} \text{ and } b_i = \lambda_{\text{anc}} \frac{\nu_i}{\sum_i \nu_i} \end{aligned} \tag{19}$$

The stationary point of the gradient $\hat{p}_i^\star$ in (19) is calculated in (20).

$$\hat{p}_i^\star = \frac{a_i \, y_i^{\text{reach}} + b_i \, y_i^{\text{target}}}{a_i + b_i}. \tag{20}$$

When the predicted status agrees with the label, and $y_i^{\text{reach}} = y_i^{\text{target}}$, then $\hat{p}_i^\star$ equals that common label and the two terms in (19) are aggregated as in (21), thereby accelerating the model's learning of correctly energised predictions.

$$\hat{p}_i^\star \;\approx\; \frac{\lambda_{\text{reach}} \, \omega_i \, y_i^{\text{reach}} + \lambda_{\text{anc}} \, \nu_i \, y_i^{\text{target}}}{\lambda_{\text{reach}} \, \omega_i + \lambda_{\text{anc}} \, \nu_i} \tag{21}$$

When the predicted status disagrees with the label, the confidence factor γ applies the class weight ω_i^{cls} to the loss term that predicts 0 at that bus. Whenever the topology label predicts 0, the model is driven by the primary BCE based on the topology, and when the labelled status predicts 0, the model is driven by the secondary BCE anchor.

1) When $y_i^{\text{reach}} = 0$ and $y_i^{\text{target}} = 1$: The topology label is de-energized $\omega_i^{\text{cls}} = \omega_{\text{off}}$ while the labelled bus status is $\nu_i = 1$ so (20) reduces to (22).

$$\hat{p}_i^\star \approx \frac{\lambda_{\text{anc}}}{\lambda_{\text{reach}} \, w_{\text{off}} + \lambda_{\text{anc}}} = \frac{0.25}{w_{\text{off}} + 0.25} \tag{22}$$

In this case, the model predicts the bus to be de-energised while the bus label says it is energised, so ω_{off} is anchored to the class weight of the primary loss term to drive the gradient passed to the model.

2) When $y_i^{\text{reach}} = 1$ and $y_i^{\text{target}} = 0$: The topology label is energized $\omega_i^{\text{cls}} = 1$ while the labelled bus status is $\nu_i = \omega_{\text{off}}$ so (20) reduces to (23).

$$\hat{p}_i^\star \approx \frac{\lambda_{\text{reach}}}{\lambda_{\text{reach}} + \lambda_{\text{anc}} \, w_{\text{off}}} = \frac{1}{1 + 0.25 \, w_{\text{off}}} \tag{23}$$

Conversely, when the model predicts the bus to be energised, but the label is de-energised, ω_{off} is anchored to the weight of the secondary loss for the labelled bus that predicts 0, thereby driving the gradient passed to the model.

B. Enforcing physical topology on voltage magnitude

The voltage head is trained using the voltage terms $\mathcal{L}_{\text{on}}$ and $\mathcal{L}_{\text{off}}$ to ensure the model's voltage predictions align with the network's physical topology, as derived from the primary loss. The voltage head is supervised on only the PQ and REF buses that receive the model output. Enforcing the physical topology on the power system quantities accelerates training of the model's head using other specialised tasks. The gradient passed to the model is split between buses predicted to be energised ($y_i^{\text{reach}} = 1$) and those predicted to be de-energised ($y_i^{\text{reach}} = 0$).

1) De-energised buses: For buses predicted to be de-energised, only the gradient from $\mathcal{L}_{\text{off}}$ contributes and is calculated in (24). The de-energised class is weighted by only the de-energised buses n_{off} to ensure that the gradient is not diluted by buses predicted to be energised that are the majority in the dataset, thereby making it a stronger penalty for the model to learn compared to the energised buses class with a larger η_j.

$$\frac{\partial \mathcal{L}_{\text{VLD}}}{\partial \hat{V}_i} = \frac{\lambda_{\text{off}}}{n_{\text{off}}} \left(2\hat{V}_i + \lambda_1 \, \text{sgn}(\hat{V}_i) \right) \tag{24}$$

In this first term, the gradient forces the restoration of the prediction proportional to $\hat{V}_i$, so that the model reinforces its prediction probability from other losses that contribute gradients to the model's voltage head during training. This approach ensures that the loss function can be efficiently used to guide the model's awareness of the network topology while reinforcing voltage predictions. The second term adds a *constant* pull of magnitude $\lambda_1 \lambda_{\text{off}} / m_U$ toward zero with a geometric decay so that the predicted voltage is forced towards zero without perturbing already a predicted zero voltage to ensure consistency with the derived de-energised topology.

2) Energised buses: Buses predicted to be energised only contribute the gradient from $\mathcal{L}_{\text{on}}$ calculated in (25).

$$\lambda_{\text{on}} \frac{2 \, \text{sg}(\hat{p}_i)}{\sum_j \eta_j} \left(\hat{V}_i - V_i \right) \tag{25}$$

The voltage predictions for buses already predicted as energised are guided by the regression MSE gradient scaled by the model's current confidence in its voltage predictions. This term ensures that the model focuses on voltage supervision for other tasks on buses predicted to be energised. The voltage predictions

V. Evaluation of Prediction Accuracy

To evaluate the accuracy of the model in predicting the bus status logit, classification metrics of OFF-class precision, recall and F1 are computed by evaluating the number of True Positives (TP) where $\hat{p}_i < \theta \wedge s_i = 0$, False Positives (FP) where $\hat{p}_i < \theta \wedge s_i = 1$ and False Negatives (FN) where $\hat{p}_i \geq \theta \wedge s_i = 0$ as in (26).

$$\begin{aligned} &\text{OFF Recall } = \frac{TP}{TP + FN}, \text{ Off precision } = \frac{TP}{TP + FP} \\ &\text{and F1} = \frac{2 \times \text{Off Precision} \times \text{Off Recall}}{\text{Off Precision} + \text{Off Recall}} \end{aligned} \tag{26}$$

The mean average error (MAE) on predicted bus voltages for de-energised buses and Root Mean Square Error (RMSE) for energised buses can also be used to evaluate the model's performance, with the former expected to tend to zero as the model becomes more accurate at enforcing network topology for voltage predictions of de-energised buses and the latter expected to reduce as the model's accuracy of voltage predictions for energised buses improves from other loss functions contributing gradient to the model's voltage head.

VI. Results

The model was trained using the loss function and used to make zero-shot predictions on the radial IEEE 33-bus network with randomised N-3 contingencies, with results shown in Fig. 2

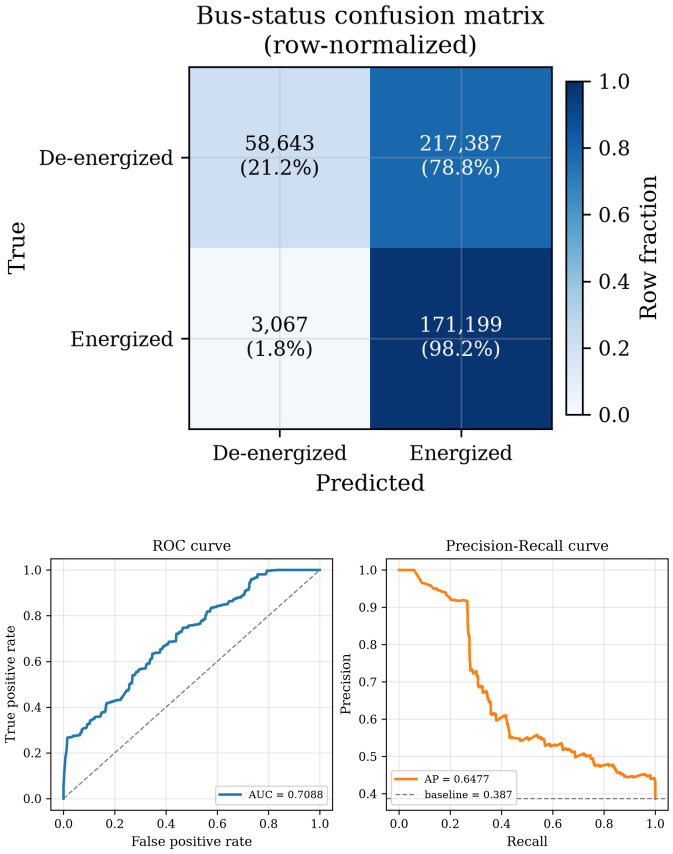


Fig. 2. Results from zero-shot predictions

VII. Conclusion

Physical connectivity to a voltage source is a key physical constraint for power system buses to remain energised. A loss function has been developed in this letter to train a grid foundation model on the network topology. The self-supervised loss passes gradients to a graph transformer that predicts the probability of a bus being energised following a network contingency, and enforces topological awareness in voltage predictions by driving voltages on buses labelled as de-energised towards zero. This differentiable loss can be integrated in the pre-training of Grid Foundation models for distribution power systems that have rampant failures and real-time network reconfiguration.

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